

Implementing Agentic RAG using Spring AI

JPrime 2026 Workshop

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Implementing Agentic RAG using Spring AI

Agenda

- What is RAG
- Methods of RAG improvements
- Agentic RAG
- Quick Demo
- Practical RAG implementation using Spring AI

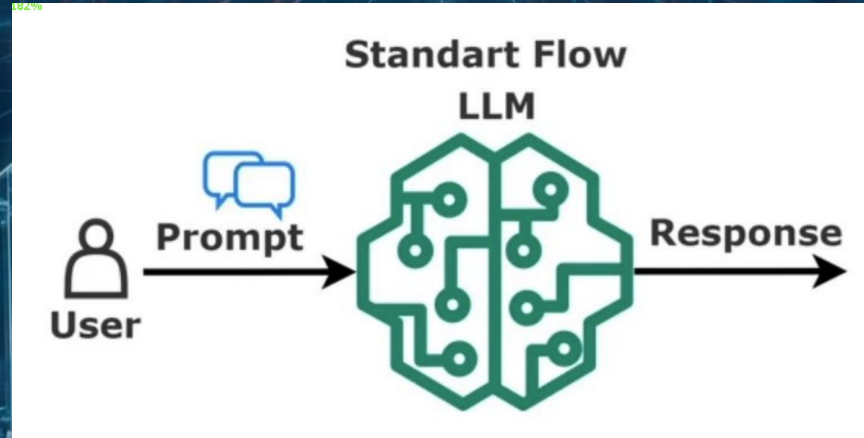
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What is RAG

- RAG is a technique that combines an LLM with an external, dynamic knowledge source and enhances its ability to generate more accurate and relevant responses

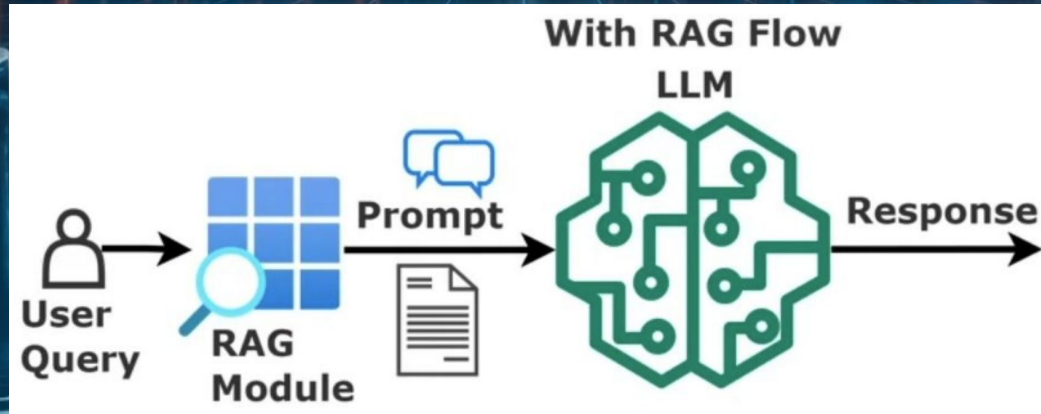
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What is RAG



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What is RAG



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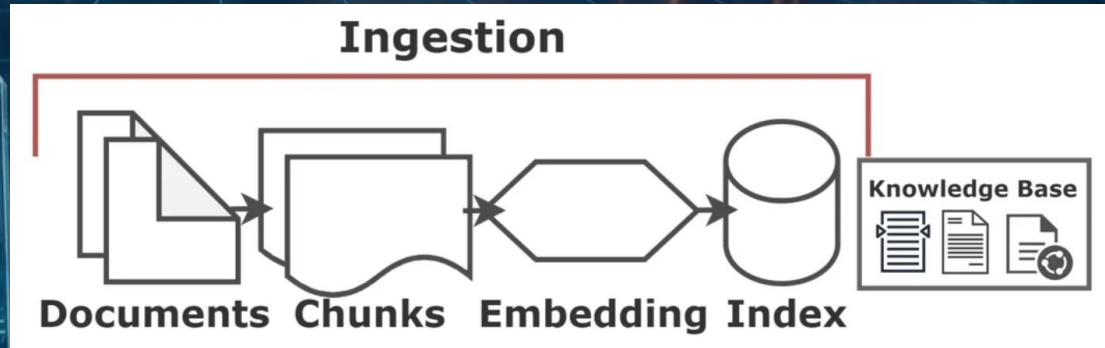
What is RAG

- The RAG process is divided into two major stages
 - Stage 1: Ingestion Phase
 - Stage 2: Real Time Query Processing - RAG Execution
- Usually they are separated by time and execution logic

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What is RAG

- Ingestion Stage
 - Documents loading
 - Chunking
 - Embedding
 - Indexing



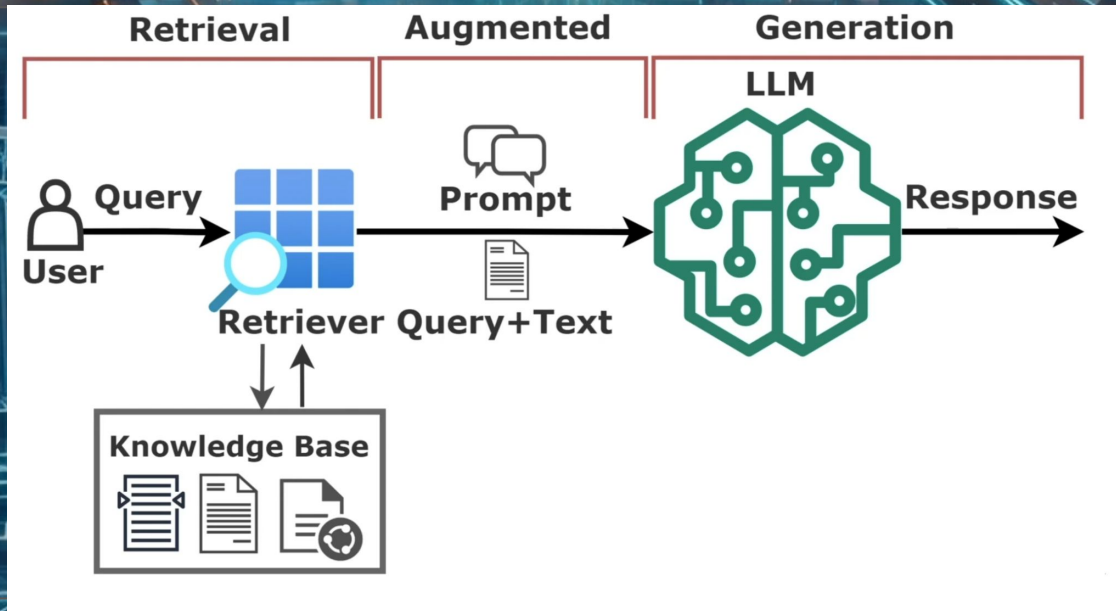
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What is RAG

- Real-Time Query Processing Stage
 - The process proceeds through three sequential and logical phases:
 - **Retrieval, Augmentation, and Generation**
 - Generates an accurate and contextually supported response from the LLM, overcoming its knowledge limitations

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Methods of RAG improvements

- Contextual retrieval

- Add contextual info about every chunk - summary of the chunk
- Reduces retrieval failures - chunks are self contained
- Expensive, slower ingestion - for every chunk there must be

LLM call

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Methods of RAG improvements

- Query Expansion
 - Uses the LLM to enrich the query
 - Expand the query into more detailed one
 - Improved retrieval precision by adding relevant context and specific information

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Methods of RAG improvements

- Multi-Query RAG
 - With the help of LLM creates several query variations
 - Run all searches to receive different results
 - Better data coverage
 - Slower

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Methods of RAG improvements

- Reranking
 - Two stages of retrieval: Vector search and Reranking results with the help of the model
 - Significantly better precision
 - Slower than pure vector search

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Methods of RAG improvements

- Agentic RAG
 - Agent autonomously chooses between multiply retrieval tools
 - Collect data between different data sources
 - Better result for Retrieving phase
 - Complex, slower and sometimes unpredictable

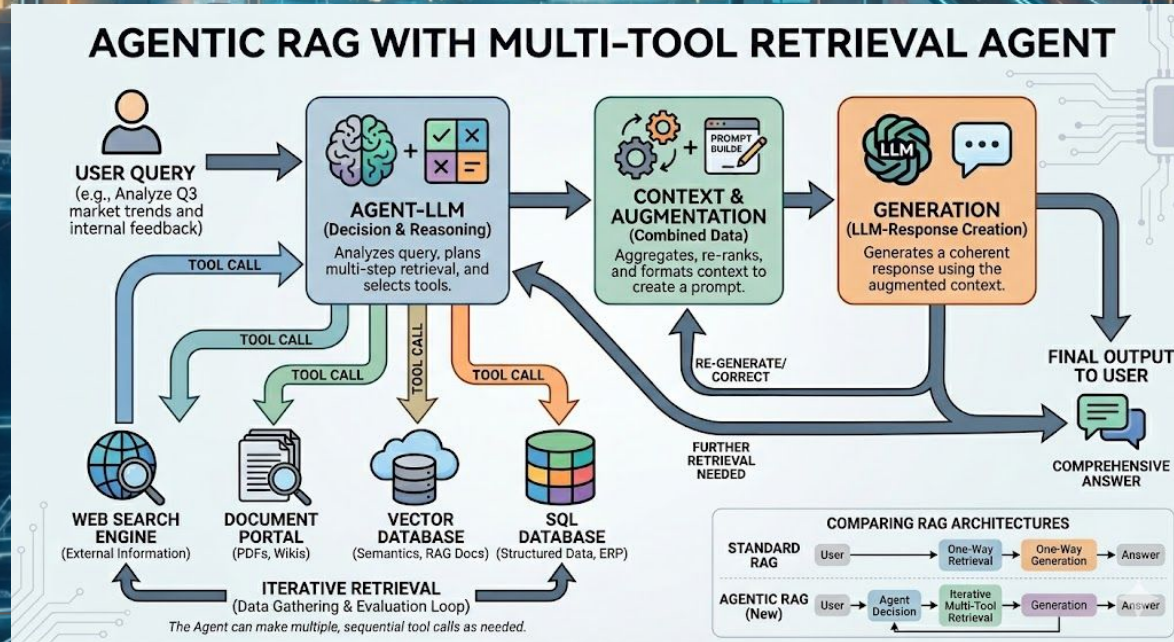
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Agentic RAG



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Thank you!

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